Probability Rules & Conditional Probability

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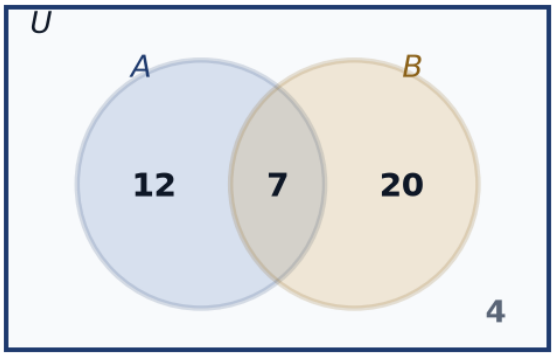
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Basic probability rules

A fair six-sided die is rolled. Find the probability the roll is even, and the probability the roll is greater than 4.

2 [Foundational] • Target: Basic probability rules

For events A and B, $P(A) = 0.4$, $P(B) = 0.3$, and $P(A \cup B) = 0.6$. Find $P(A \cap B)$ using the addition rule.



3 [Foundational] • Target: Mutually exclusive and independent events

Explain the difference between mutually exclusive events and independent events, and give a brief example of each.

4 [Foundational] • Target: Mutually exclusive and independent events

Two events A and B satisfy $P(A) = 0.5$ and $P(B) = 0.5$, and they are mutually exclusive. Explain why A and B cannot also be independent (unless one has probability 0).

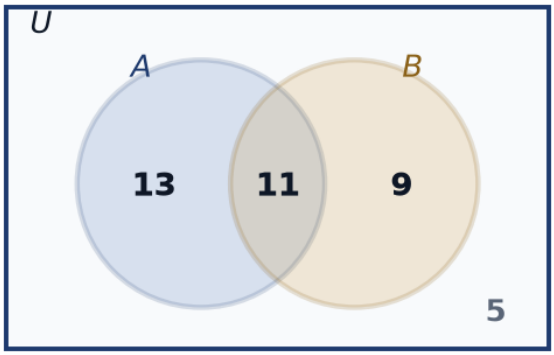
5 [Foundational] • Target: Conditional probability

In a class, the probability a student likes math is 0.6, and the probability a student likes both math and science is 0.42. Find the probability a student likes science given that they like math.

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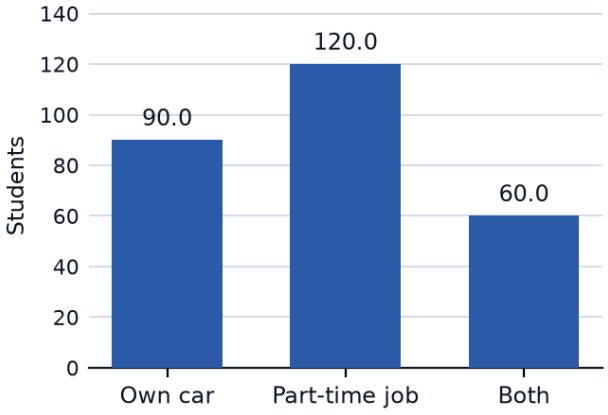
6 [Foundational] • Target: Conditional probability

Use the multiplication rule to find $P(A \cap B)$ if $P(A) = 0.5$ and $P(B \mid A) = 0.4$.



7 [Foundational] • Target: Two-way tables and conditional probability

A survey of 200 students finds 90 own a car, 120 have a part-time job, and 60 both own a car and have a job. Find the probability a randomly selected student has a job, given that they own a car.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Two-way tables and conditional probability

Using the same survey data, determine whether owning a car and having a job are independent events, showing your reasoning.

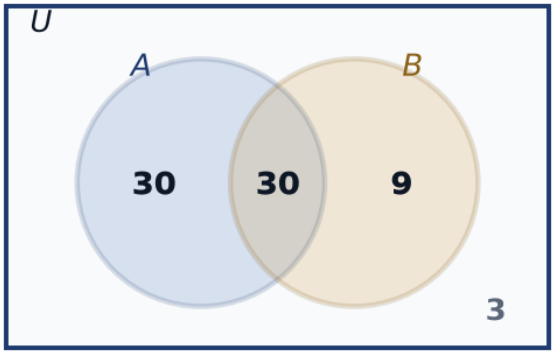
9 [Intermediate] • Target: Tree diagrams and sequential events

A factory has two machines: Machine A produces 60% of items with a 2% defect rate, and Machine B produces 40% of items with a 5% defect rate. Find the probability a randomly selected item is defective, using the law of total probability.

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10 [Intermediate] • Target: Applying probability rules to real contexts

A weather forecaster states there is a 30% chance of rain on Saturday and a 30% chance of rain on Sunday, and treats the two days as independent. Find the probability it rains on both days, and the probability it rains on at least one of the two days.



11 [Intermediate] • Target: The complement rule

A box of 50 light bulbs contains 6 defective bulbs. If one bulb is selected at random, find the probability it is NOT defective, using the complement rule.

12 [Intermediate] • Target: The complement rule

A student estimates there is a 65% chance of rain at least once during a 3-day trip, assuming each day is independent with the same daily rain probability p. Find p, to three decimal places, using $P(\text{no rain all 3 days}) = (1 - p)^3$.

13 [Intermediate] • Target: Combining independence and conditional probability

Two lie detector tests are given to a suspect, each correctly detecting deception 90% of the time (independently of each other). If the suspect is actually lying, find the probability both tests correctly detect it.

14 [Intermediate] • Target: Combining independence and conditional probability

A bag contains 5 red marbles and 3 blue marbles. Two marbles are drawn without replacement. Find the probability both are red.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

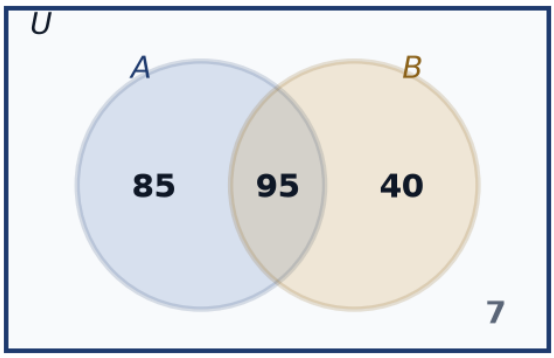
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15 [Challenging] • Target: Combining independence and conditional probability • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

At a certain hospital, 40% of patients admitted have condition A, and 25% have condition B. Additionally, 10% of all patients have both conditions A and B. Draw on your understanding of the addition rule to find the probability a randomly selected patient has condition A or condition B (or both), and separately explain why simply adding $P(A) + P(B)$ without adjustment would overcount certain patients.

16 [Challenging] • Target: Combining independence and conditional probability • Cross-Lesson: 1.01 Measures of Center & Spread

An airline reports that 85% of its flights depart on time. Among on-time flights, 95% also arrive on time, while among delayed departures, only 40% arrive on time. Using a tree-diagram approach and the law of total probability, find the overall probability that a randomly selected flight arrives on time.



17 [Challenging] • Target: Combining independence and conditional probability • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A quality inspector at a factory knows that 3% of all items produced are defective. The inspection process correctly identifies a defective item 98% of the time, but also incorrectly flags a non-defective item as defective 2% of the time (a false positive). If an item is flagged as defective by the inspection process, use the given information and the multiplication rule to find the probability that the item actually IS defective, showing every step of your reasoning (this requires combining the law of total probability with conditional probability, similar to Bayes' theorem).

18 [Challenging] • Target: Combining independence and conditional probability • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A survey of 500 adults finds that 280 exercise regularly, 220 eat a balanced diet, and 150 do both. Using this information, first find the probability a randomly selected adult exercises regularly or eats a balanced diet (or both), and then find the probability an adult eats a balanced diet given that they exercise regularly, showing every step of your work clearly.

19 [Challenging] • Target: Conditional probability • Cross-Lesson: 2.01 Sampling Methods & Bias

Level 3 · AdvancedUsing the student prep table (Studied: 54 passed, 6 failed; Did not study: 18 passed, 22 failed; Total passed: 72 out of 100), are the events 'passed' and 'studied' independent?

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20 [Challenging] • Target: Two-way tables and conditional probability • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

Events A and B satisfy $P(A) = 0.40$, $P(B) = 0.50$, and $P(A \cup B) = 0.70$. (a) Find $P(A \cap B)$. (b) Determine whether A and B are independent. (c) Determine whether A and B are mutually exclusive.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Basic probability rules

A bag contains 7 red marbles and 6 blue marbles. Two marbles are drawn without replacement. Find the probability both are blue.

a. $\frac{19}{65}$

b. $\frac{6}{65}$

c. $\frac{6}{13}$

d. $\frac{5}{26}$

22 [Foundational] • Target: Mutually exclusive and independent events

A factory has two machines: Machine A produces 60% of items with a 10% defect rate, and Machine B produces 40% of items with a 15% defect rate. Letting D denote the event that an item is defective, find $P(D)$ for a randomly selected item, using the law of total probability.

a. 12.00%

b. 13.00%

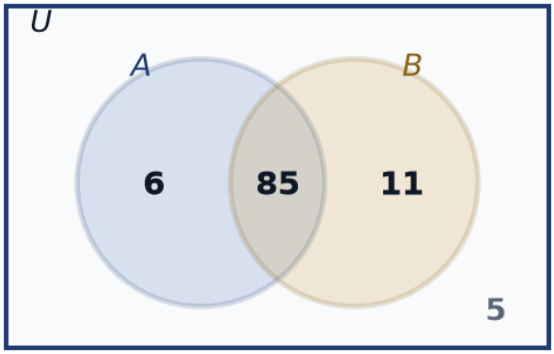
c. 11.50%

d. 12.50%

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23 [Foundational] • Target: Conditional probability

A quality inspector knows 6% of all items produced are defective (event D). The inspection process correctly flags a defective item 85% of the time, but also flags a non-defective item as defective 1% of the time (a false positive). If an item is flagged as defective (event F), find $P(D \mid F)$, to one decimal place.



a. 85.0%

b. 81.4%

c. 87.4%

d. 84.4%

24 [Foundational] • Target: Two-way tables and conditional probability

If two events A and B are mutually exclusive, and $P(A) = 0.35$ and $P(B) = 0.45$, what is $P(A \text{ and } B)$?

a. 0

b. 0.1575

c. 0.80

d. 1.00

25 [Foundational] • Target: Tree diagrams and sequential events

A survey of 159 employees found that 4 are in the Marketing department and working remotely, 26 are in the Marketing department and working on-site, 88 are not in the Marketing department and working remotely, and 41 are not in the Marketing department and working on-site. Based on this data, are the events "in the Marketing department" and "working remotely" independent?

a. No, because $P(\text{in the Marketing department}) = \frac{30}{159}$ but $P(\text{in the Marketing department} \mid \text{working remotely}) = \frac{4}{92}$, which are not equal

b. Yes, because the total number of employees is greater than 100

c. No, because $P(\text{in the Marketing department})$ cannot be computed from this table

d. Yes, because both categories have positive counts in every cell

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26 [Foundational] • Target: Applying probability rules to real contexts

A survey of 177 patients found that 80 are receiving the treatment and showing improvement, 32 are receiving the treatment and showing no improvement, 50 are receiving the placebo and showing improvement, and 15 are receiving the placebo and showing no improvement. Based on this data, are the events "receiving the treatment" and "showing improvement" independent?

a. Yes, because the total number of patients is greater than 100

b. No, because $P(\text{receiving the treatment})$ cannot be computed from this table

c. No, because $P(\text{receiving the treatment}) = \frac{112}{177}$ but $P(\text{receiving the treatment} \mid \text{showing improvement}) = \frac{80}{130}$, which are not equal

d. Yes, because both categories have positive counts in every cell

27 [Foundational] • Target: The complement rule

A survey of 162 students found that 20 are in the Science Club and playing a varsity sport, 36 are in the Science Club and not playing a varsity sport, 80 are not in the Science Club and playing a varsity sport, and 26 are not in the Science Club and not playing a varsity sport. Based on this data, are the events "in the Science Club" and "playing a varsity sport" independent?

a. No, because $P(\text{in the Science Club}) = \frac{56}{162}$ but $P(\text{in the Science Club} \mid \text{playing a varsity sport}) = \frac{20}{100}$, which are not equal

b. Yes, because both categories have positive counts in every cell

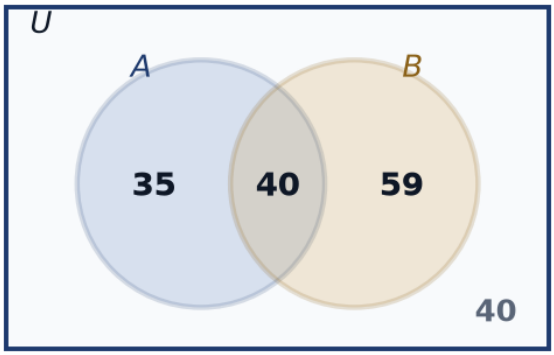
c. Yes, because the total number of students is greater than 100

d. No, because $P(\text{in the Science Club})$ cannot be computed from this table

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28 [Foundational] • Target: Combining independence and conditional probability

A survey of 183 voters found that 35 are under 40 years old and supporting the proposal, 59 are under 40 years old and opposing the proposal, 46 are 40 years old or older and supporting the proposal, and 43 are 40 years old or older and opposing the proposal. Based on this data, are the events "under 40 years old" and "supporting the proposal" independent?



a. No, because $P(\text{under 40 years old}) = \frac{94}{183}$ but $P(\text{under 40 years old} \mid \text{supporting the proposal}) = \frac{35}{81}$, which are not equal

b. Yes, because the total number of voters is greater than 100

c. Yes, because both categories have positive counts in every cell

d. No, because $P(\text{under 40 years old})$ cannot be computed from this table

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Basic probability rules

The table shows voting results for a proposed park expansion: DistrictVoted YesVoted NoTotalNorth District12080200South District180120300Total300200500 Are the events 'Voted Yes' and 'Resides in North District' independent?

a. Yes, because $P(\text{Yes} \mid \text{North}) = P(\text{Yes}) = 0.60$

b. No, because more people voted yes in the South District

c. No, because $120 eq 180$

d. Cannot be determined without a larger sample

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30 [Intermediate] • Target: Mutually exclusive and independent events

A survey of 157 shoppers found that 57 are using a coupon and spending over \$50, $14x50$ or less, 33 are not using a coupon and spending over \$50, $and53x50$ or less. Based on this data, are the events "using a coupon" and "spending over \$50" independent?

a. Yes, because both categories have positive counts in every cell

b. No, because $P(\text{using a coupon})$ cannot be computed from this table

c. Yes, because the total number of shoppers is greater than 100

d. No, because $P(\text{using a coupon}) = \frac{71}{157}$ but $x50\})=\dfrac{57}{90}$, which are not equal

31 [Intermediate] • Target: Conditional probability

Two events E and F satisfy $P(E) = 0.5$, $P(F) = 0.3$, and $P(E \cup F) = 0.65$. Are E and F independent?

a. Yes, because $P(E \cap F) = P(E)P(F) = 0.15$

b. No, because $P(E \cap F) eq 0.15$

c. No, because $P(E) eq P(F)$

d. Cannot be determined

32 [Intermediate] • Target: Two-way tables and conditional probability

If $P(A) = 0.4$ and $P(B) = 0.3$, and A and B are mutually exclusive, what is $P(A \mid B)$?

a. 0

b. 0.12

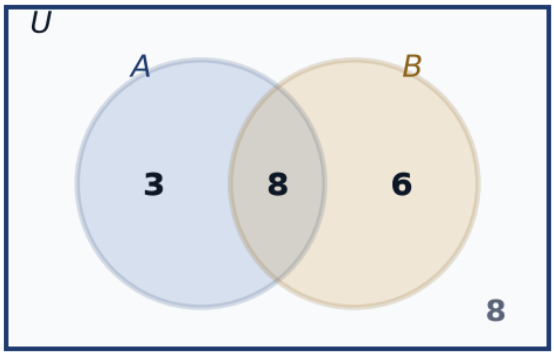
c. 0.40

d. Undefined

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33 [Intermediate] • Target: Tree diagrams and sequential events

Events A and B are independent, with $P(A) = \frac{3}{8}$ and $P(B) = \frac{1}{6}$. What is $P(A \text{ and } B)$?



a. 19/48

b. 1/16

c. 9/80

d. 5/16

34 [Intermediate] • Target: Applying probability rules to real contexts

Two events A and B are independent. If $P(A) = 0.4$ and $P(B) = 0.7$, what is $P(A \text{ and } B)$?

a. 0.28

b. 0.30

c. 0.55

d. 0.82

35 [Intermediate] • Target: The complement rule

If events A and B are mutually exclusive:

a. $P(A \cap B) = P(A)P(B)$

b. $P(A \cup B) = P(A) + P(B)$

c. They are independent

d. $P(A \cap B) > 0$

36 [Intermediate] • Target: Combining independence and conditional probability

Three independent alarm systems each have a 90% chance of functioning during an emergency. What is the probability that AT LEAST ONE system functions during an emergency?

a. 0.900

b. 0.970

c. 0.999

d. 1.000

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37 [Intermediate] • Target: Basic probability rules

If $P(A) = 0.4$, $P(B) = 0.5$, and A and B are independent, what is $P(A \text{ or } B)$?

a. 0.70

b. 0.75

c. 0.80

d. 0.90

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Mutually exclusive and independent events • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A national health organization is investigating the relationship between daily screen time (in hours) and the number of hours of sleep reported by adults. They collect data from a sample of 1200 adults. The data is summarized in a two-way table, where the row totals represent categories of daily screen time and the column totals represent categories of reported sleep hours. The organization is particularly interested in the cells where adults report less than 4 hours of screen time and get 7 or more hours of sleep. If the overall row total for 'less than 4 hours of screen time' is 250, and the overall column total for '7 or more hours of sleep' is 300, what is the expected count for the cell representing adults with less than 4 hours of screen time who get 7 or more hours of sleep, assuming independence between screen time and sleep hours?

a. 62.50

b. 75.00

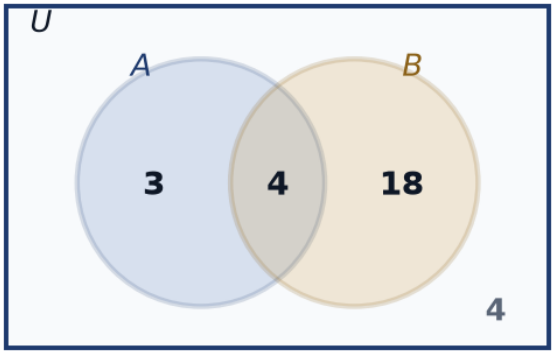
c. 83.33

d. 250.00

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39 [Challenging] • Target: Conditional probability • Cross-Lesson: 2.01 Sampling Methods & Bias

A pharmaceutical company is testing a new drug designed to reduce blood pressure. They decide to conduct a randomized controlled experiment. To investigate the potential for confounding variables related to lifestyle, they collect preliminary data on participants' self-reported levels of physical activity and dietary habits. They categorize physical activity into three levels: 'Low', 'Moderate', and 'High'. Dietary habits are categorized into four types: 'Vegetarian', 'Pescatarian', 'Low-Carb', and 'Omnivore'. If the company wants to ensure that any observed association between these two categorical variables in their sample data can be rigorously tested for statistical significance using a chi-square test of independence, and they obtain a contingency table with 3 rows (for physical activity) and 4 columns (for dietary habits), what is the maximum number of independent pieces of information contributing to the chi-square statistic, often referred to as the degrees of freedom, in this specific test?



a. 12

b. 9

c. 6

d. 2

40 [Challenging] • Target: Two-way tables and conditional probability • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A manufacturing company is testing a new quality control process. They have collected data on the number of defects found in batches of a product, categorized by the production shift. The data is summarized in the two-way table below: | Shift | Defects Found | No Defects Found | Total | |---|---|---|---| | Morning | 15 | 85 | 100 | | Afternoon | 25 | 75 | 100 | | Evening | 10 | 90 | 100 | | **Total** | **50** | **250** | **300** | If a test of independence is to be performed on this data, what is the expected number of defects found in the afternoon shift?

a. 20.00

b. 16.67

c. 25.00

d. 8.33

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41 [Challenging] • Target: Tree diagrams and sequential events • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A researcher is conducting a survey on the relationship between college major and career satisfaction. They collected data from 150 students and organized it into a two-way table. If the number of rows in the table represents 3 different college majors and the number of columns represents 5 different levels of career satisfaction, what is the df for a chi-square test of independence?

a. 12

b. 8

c. 2

d. 4

42 [Challenging] • Target: Applying probability rules to real contexts • Cross-Lesson: 3.03 The Normal Distribution

A researcher is conducting a survey to understand the relationship between two categorical variables: preferred social media platform and year in school. The data collected is summarized in a two-way table. If the table has 4 rows and 3 columns, what is the degrees of freedom for a chi-square test of independence performed on this table?

a. 5

b. 6

c. 2

d. 12

43 [Challenging] • Target: The complement rule • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

Events A and B are independent with $P(A) = 0.33$ and $P(B) = 0.88$. Find $P(A \cap B)$.

a. 0.4904

b. 0.2904

c. 0.2404

d. 0.1904

44 [Challenging] • Target: Combining independence and conditional probability • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A survey was conducted among 200 college students regarding their preferred modes of transportation for commuting to campus. The results are summarized in the two-way table below. | | Car | Bicycle | Public Transit | Total | |----------------|-----|---------|----------------|-------| | **Freshman** | 45 | 20 | 35 | 100 | | **Sophomore** | 30 | 25 | 45 | 100 | | **Total** | 75 | 45 | 80 | 200 | If a student is chosen at random from those who prefer to commute by bicycle, what is the probability that the student is a freshman? Express your answer as a decimal rounded to the nearest hundredth.

a. 0.44

b. 0.50

c. 0.56

d. 0.67

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45 [Challenging] • Target: Basic probability rules • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A survey was conducted on a group of 100 students regarding their preferred mode of transportation to school. The results are summarized in the two-way table below. If a student is chosen at random from this group, and it is known that the student prefers cycling, what is the probability that the student is also a member of the chess club? The value of x represents the number of students who prefer public transport and are not in the chess club. | | Chess Club | Not Chess Club | Total | | :---------- | :--------: | :----------: | :---: | | Cycling | 15 | x | 30 | | Public Trans| 20 | 55 | 75 | | Total | 35 | 60 | 100 |

a. $\frac{1}{4}$

b. $\frac{3}{7}$

c. $\frac{1}{2}$

d. $\frac{5}{12}$

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